Day 8 - Factorial of a Number

Question

Ek number **N** diya gaya hai.

Uska factorial find karo.

Formula

```
n! = n × (n-1) × (n-2) × ... × 1
```

Example

```
5! = 5 × 4 × 3 × 2 × 1
```

```
= 120
```

Approach 1 - Using For Loop □

Logic

1 se start karke multiplication karte jao.

Dry Run

```
n = 5

fact = 1

fact = 1 × 1 = 1
fact = 1 × 2 = 2
fact = 2 × 3 = 6
fact = 6 × 4 = 24
fact = 24 × 5 = 120
```

Java Code

```
public class Day8 {
    public static void main(String[] args) {

        int n = 5;
        long fact = 1;

        for(int i = 1; i <= n; i++) {
            fact *= i;
        }

        System.out.println(fact);
    }
}
```

Time Complexity

```
O(N)
```

Space Complexity

```
O(1)
```

Approach 2 - Using While Loop

Java Code

```
public class Day8 {
    public static void main(String[] args) {

        int n = 5;
        long fact = 1;

        while(n > 0) {
            fact *= n;
            n--;
        }

        System.out.println(fact);
    }
}
```

Time Complexity

```
O(N)
```

Space Complexity

```
O(1)
```

Approach 3 - Using Recursion

Logic

```
fact(n) = n × fact(n-1)
```

Dry Run

```
fact(5)

5 × fact(4)

5 × 4 × fact(3)

5 × 4 × 3 × fact(2)

5 × 4 × 3 × 2 × fact(1)

5 × 4 × 3 × 2 × 1

= 120
```

Java Code

```
public class Day8 {

    static long factorial(int n) {

        if(n == 0 || n == 1)
            return 1;

        return n * factorial(n - 1);
    }

    public static void main(String[] args) {

        System.out.println(factorial(5));
    }
}
```

Time Complexity

```
O(N)
```

Space Complexity

```
O(N)
```

(Call Stack)

Approach 4 - Using BigInteger □

Problem

```
20! = 2432902008176640000
```

Ye long ki limit ke paas hai.

```
50!
100!
500!
```

long me store nahi honge.

Solution

Java ka BigInteger use karo.

Java Code

```
import java.math.BigInteger;

public class Day8 {

    public static void main(String[] args) {

        int n = 50;

        BigInteger fact = BigInteger.ONE;

        for(int i = 1; i <= n; i++) {
            fact = fact.multiply(BigInteger.valueOf(i));
        }

        System.out.println(fact);
    }
}
```

Interview Follow-Up Questions

Q1. 0! kitna hota hai?

```
0! = 1
```

Ye mathematical definition hai.

Q2. 1! kitna hota hai?

```
1! = 1
```

Q3. Factorial Negative Number ka hota hai?

```
Nahi
```

Normal factorial sirf:

```
0, 1, 2, 3, ...
```

ke liye defined hota hai.

Q4. Why long instead of int?

```
5! = 120
10! = 3628800
13! = 6227020800
```

13! int me fit nahi hota.

Isliye:

```
long fact
```

use karte hain.

Interview Concept of the Day

Recursion Tree

```
factorial(5)

|
5 × factorial(4)

|
4 × factorial(3)

|
3 × factorial(2)

|
2 × factorial(1)

|
1
```

Ye recursion tree ka basic form hai.

Future me:

- Recursion
- Backtracking
- Trees
- Dynamic Programming

sab me use hoga.

Summary

Approach	Time	Space
For Loop □	O(N)	O(1)
While Loop	O(N)	O(1)
Recursion	O(N)	O(N)
BigInteger	O(N)	O(1)

Practice Questions

Find:

1

```
3!
```

2

```
4!
```

3

```
6!
```

4

```
7!
```

5

```
10!
```